Republic of Tunisia
National Defense Ministry
Army
Military Academy
Directorate of Education and Training
Department of Higher Education

END OF STUDIES PROJECT REPORT (PFE)

QUANTUM-INSPIRED FEDERATED LEARNING IDS FOR IOT/WSN

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Academic Year: 2025 / 2026

Acknowledgements

Abstract

À compléter.

Résumé

À compléter.

Contents

Acknowledgements **ii**

Abstract **iii**

Résumé **iv**

List of Figures **vii**

List of Tables **viii**

Acronyms **ix**

General Introduction **1**

1 State of the Art **2**

I IoT/WSN & Fog/Edge Computing 2

I.1 Constraints 2

I.2 Offloading & Orchestration 2

II Intrusion Detection Systems for IoT/WSN 2

II.1 Detection Approaches 2

II.2 Deep Learning-based IDS 2

II.3 Lightweight IDS 2

II.4 Challenges and Limitations 2

III Federated Learning for IoT Network Security 2

III.1 Context and Motivation 2

III.2 Definition and Positioning of Federated Learning 3

III.3 Mathematical Formulation of Federated Learning 3

III.4 Objective Function and GTV-Based Federated Learning Formulation 5

IV Federated Learning Algorithms: Families, Types, and Trade-offs 7

IV.1 Centralized Federated Learning Algorithms 7

IV.1.1 Representative Algorithms 7

IV.1.2 Advantages and Limitations 8

IV.2 Decentralized Federated Learning Algorithms 8

IV.2.1 Key Principles 8
IV.2.2 Advantages and Limitations 8
IV.3 Synchronous versus Asynchronous Federated Learning 9
IV.3.1 Synchronous Federated Learning 9
IV.3.2 Asynchronous Federated Learning 9
IV.4 Federated Learning Flavours and Their Objectives 9
IV.4.1 Single-domain Federated Learning 9
IV.4.2 Horizontal Federated Learning (HFL) 9
IV.4.3 Vertical Federated Learning (VFL) 9
IV.4.4 Cross-silo Federated Learning 9
IV.4.5 Cross-device Federated Learning 10
IV.4.6 Personalized Federated Learning (PFL) 10
IV.5 Differential Privacy in Federated Learning 10
IV.5.1 Comprehensive Comparison of Federated Learning Algorithms 12
V Quantum-Inspired Optimization and Machine Learning 12
V.1 Quantum-Inspired Adaptive Resource Management (QIARM) . . . 12
V.2 Quantum Genetic Algorithm (QGA) 12
V.3 Federated Tensor Networks (FedTN) / Matrix Product States (MPS) 12
V.4 Comparison of Approaches 12
VI Synthesis and Research Gap 12
VI.1 Identified Limitations 12
VI.2 Positioning of the Proposed Work 12

2 Requirements analysis and Design **13**
I Plan 13

3 Implementation **14**
I Plan 14

4 Results analysis and discussion **15**
I Plan 15

General Conclusion **16**

References **17**

List of Figures

1.1 Illustration of the training data exhaustion problem for Large Language Models (LLMs). 3
1.2 Graph-based representation of a federated learning system, illustrating distributed datasets, hypothesis spaces, and local loss functions across clients. 4
1.3 Federated Learning workflow illustrating client selection, local training, transmission of local updates, and global model aggregation across federated rounds. 5

List of Tables

1.1 Algorithmes d'apprentissage fédéré : caractéristiques et comparaison 11

Acronyms

IoT	Internet of Things
WSN	Wireless Sensor Networks
IDS	Intrusion Detection System
FL	Federated Learning
FedAvg	Federated Averaging
QIARM	Quantum Inspired Adaptive Resource Management
QGA	Quantum Genetic Algorithm
FedTN	Federated Tensor Networks
MPS	Matrix Product State

General Introduction

1. State of the Art

I IoT/WSN & Fog/Edge Computing

I.1 Constraints

I.2 Offloading & Orchestration

II Intrusion Detection Systems for IoT/WSN

II.1 Detection Approaches

II.2 Deep Learning-based IDS

II.3 Lightweight IDS

II.4 Challenges and Limitations

III Federated Learning for IoT Network Security

III.1 Context and Motivation

The fast pace of development of artificial intelligence models, especially **Large Language Models (LLMs)** and sophisticated detection systems, is increasingly challenged by a fundamental structural limitation: **LLMs are running out of training data**. Contemporary models require not only large volumes of data, but also **diverse, representative, and high-quality datasets** to ensure reliable generalization.

In most critical application domains—such as the Internet of Things (IoT), cybersecurity, healthcare, and industrial infrastructures—the most valuable data is inherently **private, sensitive, and distributed**. Centralized storage and processing of such data raise serious concerns related to **data privacy**, **regulatory compliance** (e.g., GDPR, HIPAA), and increased **attack surface exposure**.

In this context, **Federated Learning (FL)** has emerged as a promising paradigm that enables collaborative model training while keeping data **locally stored and undisclosed**. By design, FL enhances the performance and robustness of global models while preserving data confidentiality. Its ability to balance **learning quality**, **privacy preservation**, and

system security makes it particularly well suited for **dynamic and heterogeneous IoT environments**.

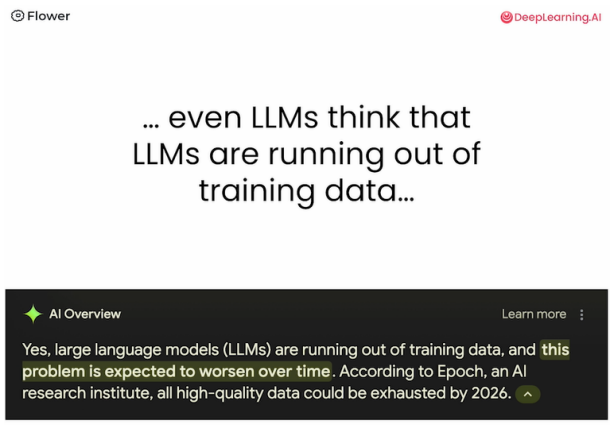

Figure 1.1: Illustration of the training data exhaustion problem for Large Language Models (LLMs).

Source: DeepLearning.AI

III.2 Definition and Positioning of Federated Learning

Federated Learning (FL) is a distributed machine learning paradigm in which multiple autonomous entities, referred to as *clients*, collaboratively train a shared model **without exchanging their raw data**. In contrast to the conventional centralized learning approach—where data is collected, stored, and processed on a single server—FL is founded on the principle of **data locality**. Each client keeps its data locally and only communicates **model updates**, such as weights, gradients, or intermediate parameters, to a global aggregation system.

From a theoretical perspective, Federated Learning does not constitute a new form of artificial intelligence. Rather, it is a **training paradigm** that can be applied to existing machine learning and deep learning models. It belongs to the broader field of distributed learning, with a particular focus on **privacy preservation**, **decentralization**, and **robustness**.

From a conceptual perspective, Federated Learning **does not constitute a new type of artificial intelligence** but rather a **training methodology** applicable to existing machine learning and deep learning models. It belongs to the broader field of **distributed learning**, with an explicit focus on privacy protection, decentralization, and system resilience.

III.3 Mathematical Formulation of Federated Learning

A Federated Learning system can be formalized as a **federated network** consisting of a set of devices denoted $\mathcal{C} = \{1, \ldots, N\}$.

Each device $i \in \mathcal{C}$:

- Possesses a local dataset $\mathcal{D}_i$
- Trains a local model $\mathcal{H}_i$
- Defines a local loss function $L_i(w)$

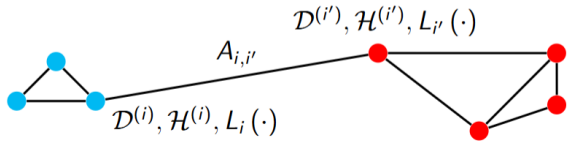

Figure 1.2: Graph-based representation of a federated learning system, illustrating distributed datasets, hypothesis spaces, and local loss functions across clients.
Source: Adapted from [1]

In an **IoT** context, each device participating in the **Federated Learning** process (sensor, gateway, or embedded node) operates in an autonomous and local manner. More specifically, each device:

- collects and stores **local data** (network traffic, system logs, operational metrics);
- trains a **local model** using its private data;
- shares only **model parameters** (weights, gradients, or intermediate updates), without ever disclosing the raw data.

The **central server**—or, more generally, the aggregation mechanism— combines the contributions of the clients in order to produce an improved **global model**. This model is then redistributed to the participating devices. The entire process is iterative and is executed over multiple successive cycles, referred to as *federated rounds*.

Illustrated federated learning process:

1. **Client selection**: at each iteration, the central aggregator selects a subset of available clients to participate in the learning process, based on criteria such as connectivity, resource availability, or device readiness;
2. **Model selection and distribution**: the current global model is selected and transmitted to the chosen clients to serve as the starting point for local training;
3. **Local training**: each client trains a local model using its privately stored data, without sharing it with the server or other clients;

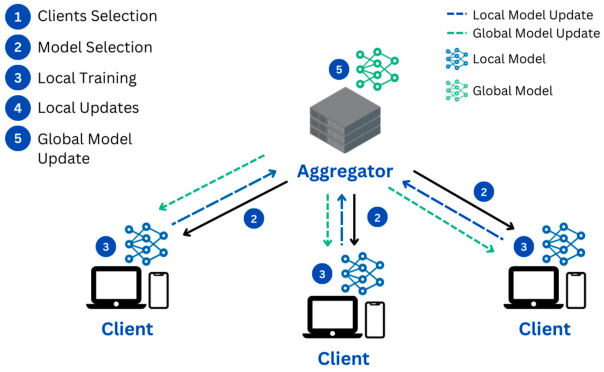

Figure 1.3: Federated Learning workflow illustrating client selection, local training, transmission of local updates, and global model aggregation across federated rounds.

4. **Transmission of local updates**: after training, clients send only the model updates (weights, gradients, or parameter differences) to the aggregator;

5. **Global model update**: the aggregator combines the received contributions from all clients, for example using the *Federated Averaging (FedAvg)* algorithm, to produce a new global model.

III.4 Objective Function and GTV-Based Federated Learning Formulation

The objective function

$$\min_{\{w_i\}} \sum_i L_i(w_i) \; + \; \alpha \sum_{(i,j)} A_{ij} \, \|w_i - w_j\|^2 \tag{1.1}$$

forms the mathematical foundation of generalized and personalized Federated Learning. It explicitly captures the fundamental trade-off between **autonomous local learning** and **controlled distributed collaboration** among the clients of a federated system.

The first term,

$$\sum_i L_i(w_i), \tag{1.2}$$

represents the local loss associated with each client. It corresponds to the classical machine learning objective, where each node (e.g., sensor, gateway, or IoT device) trains a model using its own private dataset. This term ensures that the learned models are locally adaptive to the specific data distributions of individual devices, which is particularly important in IoT environments characterized by strong heterogeneity and non-IID data. When considered alone, however, this term leads to fully independent models with no form of collaboration or knowledge sharing.

The second term,

$$\alpha \sum_{(i,j)} A_{ij} \|w_i - w_j\|^2, \tag{1.3}$$

introduces a **generalized total variation (GTV)** regularization that models collaboration between clients through a federated graph structure. The coefficients A_{ij} encode logical or functional relationships between clients, such as network topology, statistical similarity of data distributions, trust relationships, or device categories. This regularization penalizes large discrepancies between the models of neighboring clients, thereby enabling implicit knowledge sharing without requiring the exchange of raw data, which is essential for privacy preservation.

The regularization parameter $\alpha > 0$ plays a central role in balancing local autonomy and global cooperation. Small values of α favor highly personalized models with limited collaboration, whereas larger values enforce stronger similarity among client models, gradually approaching the behavior of a single shared global model. In the context of IoT security and intrusion detection systems (IDS), this parameter provides fine-grained control over the degree of security knowledge sharing, ranging from purely local detection to robust collaborative intrusion detection.

Overall, this formulation enables effective handling of data heterogeneity, mitigates local overfitting when individual datasets are small, and explicitly exploits the structural properties of IoT networks. It unifies several Federated Learning paradigms, including personalized FL, clustered FL, and global FL. From this perspective, classical approaches such as *FedAvg* can be interpreted as a special case of this framework, corresponding to a strong constraint enforcing a single shared model across all clients. In contrast, GTV minimization offers a more flexible and structure-aware federation strategy, making it particularly suitable for dynamic and distributed IoT systems.

In **synchronous Federated Learning**, global aggregation is performed only after all selected clients have completed their local training. This approach is simple and theoretically stable but suffers from the *straggler problem*, particularly critical in heterogeneous IoT networks.

Conversely, **asynchronous Federated Learning** allows clients to send their updates as soon as they are available, without waiting for others. This approach is more realistic for dynamic IoT environments, tolerates delays and disconnections, but must handle the problem of stale updates.

For IoT systems, asynchronous approaches are generally **more suitable**.

IV Federated Learning Algorithms: Families, Types, and Trade-offs

Federated Learning algorithms do not constitute a single homogeneous solution, but rather a family of learning strategies that vary along several fundamental axes. These axes directly influence their performance, robustness, and suitability for heterogeneous and resource-constrained IoT environments.

In general, Federated Learning algorithms can be categorized according to four main criteria:

- **System architecture**: centralized versus decentralized
- **Execution type**: synchronous versus asynchronous
- **Learning objective**: single global model versus personalized models
- **System robustness**: tolerance to data heterogeneity, non-IID data, faults, and security threats.

This section provides a structured overview of these algorithmic families, highlighting their underlying principles, strengths, and limitations, with particular emphasis on IoT environments and Intrusion Detection System (IDS) applications.

IV.1 Centralized Federated Learning Algorithms

In centralized Federated Learning, a federated server acts as the central coordinator of the learning process. It is responsible for initializing the global model, selecting participating clients at each federated round, aggregating local updates, and redistributing the updated model.

The learning process follows an iterative procedure:

- the server broadcasts the global model to selected clients;
- clients train the model locally using their private data;
- local updates (model parameters or gradients) are sent back to the server;
- the server aggregates these updates to produce a new global model.

IV.1.1 Representative Algorithms

FedAvg (Federated Averaging) FedAvg is the foundational Federated Learning algorithm. It computes a weighted average of local model parameters, typically proportional to the size of each client's dataset. Although simple and communication-efficient, FedAvg is particularly sensitive to data heterogeneity and non-IID data distributions.

FedProx FedProx extends FedAvg by introducing a proximal regularization term in the local optimization objective. This term limits excessive divergence between local and global models, thereby improving stability in the presence of heterogeneous data and varying client computational capabilities.

FedAdam, FedYogi, and FedAdagrad These algorithms adapt advanced optimization techniques, such as Adam and Adagrad, to the federated setting. By incorporating adaptive learning rates and momentum at the server side, they achieve faster convergence and improved robustness under noisy and heterogeneous gradients.

IV.1.2 Advantages and Limitations

Centralized FL approaches are relatively easy to deploy, well understood from a theoretical perspective, and supported by mature frameworks such as Flower and TensorFlow Federated. However, they introduce a **single point of trust and failure**, raising concerns regarding resilience, security, and scalability in large-scale IoT deployments.

IV.2 Decentralized Federated Learning Algorithms

Decentralized Federated Learning removes the central server and relies instead on **peer-to-peer communication** among participating nodes. Each client communicates only with its neighbors according to a predefined graph topology and updates its local model using both its own data and information received from neighboring models.

These approaches often rely on distributed consensus mechanisms, parameter diffusion strategies, or **Generalized Total Variation Minimization (GTVMin)**, where model similarity is enforced locally along the edges of a federated graph.

IV.2.1 Key Principles

- absence of a central coordinating server;
- local and incremental exchange of model information;
- natural adaptation to dynamic and evolving network topologies;
- improved tolerance to failures and targeted attacks.

IV.2.2 Advantages and Limitations

Decentralized approaches offer enhanced resilience and better alignment with distributed IoT systems while reducing reliance on a single coordination entity. In exchange, they typically involve higher algorithmic complexity, slower convergence, and more challenging implementation compared to centralized approaches.

IV.3 Synchronous versus Asynchronous Federated Learning

IV.3.1 Synchronous Federated Learning

In synchronous Federated Learning, global aggregation is performed only after all selected clients have completed their local training. This approach is simple to analyze and benefits from strong theoretical convergence guarantees. However, it suffers from the *straggler problem*, where slow or intermittently connected clients delay the entire learning process—an important limitation in heterogeneous IoT environments.

IV.3.2 Asynchronous Federated Learning

Asynchronous Federated Learning allows clients to transmit updates as soon as they are available, without waiting for other participants. The global model is updated incrementally and may incorporate stale updates. This approach is more realistic for dynamic IoT environments, more tolerant to disconnections and failures, and better suited for large-scale deployments, despite its increased analytical complexity.

IV.4 Federated Learning Flavours and Their Objectives

Federated Learning encompasses several variants, often referred to as *flavours*, which differ in data distribution, participant structure, and learning objectives.

IV.4.1 Single-domain Federated Learning

Clients belong to the same domain and share similar tasks and feature spaces.

Objective: learn a single global model while preserving data privacy.

IV.4.2 Horizontal Federated Learning (HFL)

Clients hold data corresponding to different entities but share the same feature space.

Objective: exploit distributed datasets with identical features and different samples.

IV.4.3 Vertical Federated Learning (VFL)

Clients share overlapping entities but possess different feature sets.

Objective: combine complementary data sources without explicit data fusion, often relying on cryptographic or secure computation techniques.

IV.4.4 Cross-silo Federated Learning

A limited number of reliable participants, such as organizations or operators, with stable infrastructure and substantial resources.

Objective: enable collaborative learning across organizational silos.

IV.4.5 Cross-device Federated Learning

A very large number of heterogeneous, intermittent, and resource-constrained IoT or mobile clients.

Objective: train a robust global model despite volatility and severe system constraints.

IV.4.6 Personalized Federated Learning (PFL)

Rather than enforcing a single shared model, Personalized FL aims to produce client-adaptive models.

Objective: combine shared global knowledge with local adaptation, which is particularly relevant for IoT-based IDS, where normal behavior varies across deployment contexts.

IV.5 Differential Privacy in Federated Learning

While Federated Learning significantly reduces the need to centralize sensitive data, model updates exchanged during the training process may still leak private information. To address this risk, **Differential Privacy (DP)** provides a rigorous and formal framework for privacy protection.

The core principle of Differential Privacy is that the presence or absence of an individual data record (e.g., the data of a specific user) should not significantly affect the output of the learning algorithm. This privacy guarantee is controlled by a parameter ε, known as the *privacy budget*:

- a **small** value of ε ensures strong privacy guarantees;
- a **large** value of ε favors higher model utility and accuracy.

In the context of Federated Learning, Differential Privacy can be enforced in two main ways:

- **Centralized Differential Privacy**, where noise is added to the aggregated model updates at the server side;
- **Local Differential Privacy**, where each client perturbs its model updates locally before transmitting them to the aggregator.

Key mechanisms for implementing Differential Privacy include **gradient clipping** to bound the sensitivity of updates, followed by the addition of **calibrated random noise**, typically drawn from Laplacian or Gaussian distributions.

Table 1.1: Algorithmes d'apprentissage fédéré : caractéristiques et comparaison

Algorithme	**Architecture / Mode / Objectif**	**Points forts**	**Limites**
FedAvg	Centralisée, synchrone, modèle global	Simple, largement adopté, référence solide	Sensible aux données non-IID, goulots d'étranglement, point de défaillance unique
FedProx	Centralisée, synchrone, modèle global robuste	Meilleure stabilité face à l'hétérogénéité, réduit la dérive client	Ajustement d'hyperparamètres supplémentaire, dépendance au serveur
FedAdam	Centralisée, synchrone, modèle global	Convergence rapide et stable, robuste aux mises à jour bruitées	Coût serveur élevé, hypothèse de confiance centralisée
FL Décentralisé (Gossip / GTV)	Décentralisée, sync/async, personnalisé ou par clusters	Pas de serveur central, tolérance aux pannes, adapté à l'IoT dynamique	Convergence lente, communication élevée, déploiement complexe
FL Asynchrone	Centralisée ou décentralisée, asynchrone, global ou personnalisé	Tolère la connectivité intermittente, évolutif	Mises à jour obsolètes, analyse de convergence difficile

IV.5.1 Comprehensive Comparison of Federated Learning Algorithms

Table 1.1 provides a comprehensive and readable comparison of representative Federated Learning algorithms, highlighting their architectural designs, execution modes, learning objectives, and main strengths and limitations in the context of heterogeneous IoT environments.

V Quantum-Inspired Optimization and Machine Learning

V.1 Quantum-Inspired Adaptive Resource Management (QIARM)

V.2 Quantum Genetic Algorithm (QGA)

V.3 Federated Tensor Networks (FedTN) / Matrix Product States (MPS)

V.4 Comparison of Approaches

VI Synthesis and Research Gap

VI.1 Identified Limitations

VI.2 Positioning of the Proposed Work

2. Requirements analysis and Design

I Plan

- À compléter.

3. Implementation

I Plan

- À compléter.

4. Results analysis and discussion

I Plan

- À compléter.

General Conclusion

References

[1] A. Jung, *Federated Learning: From Theory to Practice.* Espoo, Finland: Aalto University, 2025.